

LALIT BHAGAT

📍 Seattle, WA ✉️ lalitbhagat7@cs.ucla.edu  [lalitbhagat7](https://www.linkedin.com/in/lalitbhagat7)  [lalitbhagat7.github.io/](https://github.com/lalitbhagat7)  [Google Scholar](https://scholar.google.com/citations?user=lalitbhagat7)

EDUCATION

University of California - Los Angeles (UCLA)

Master of Science in Computer Science | GPA : 4.0/4.0

Los Angeles, CA

Sep 2021 – Dec 2023

- Awarded Graduate Diversity Fellowship (1% selection rate, full scholarship)

Jaypee Institute of Information Technology (JIIT)

B. Tech (with Honours) in Computer Science and Engineering | GPA : 8.7/10

Noida, India

July 2017 – May 2021

RESEARCH INTERESTS & EXPERTISE

Artificial Intelligence and Machine Learning, Computer Vision, Generative AI, Ranking and Recommendation Systems, Learning-to-Rank, Predictive Modeling, Representation Learning, and Large-Scale Machine Learning Systems

INDUSTRY EXPERIENCE

Amazon

Applied Scientist

Seattle, WA

Jan 2024 – Present

- Led development and launch of production ML systems across 12 countries, serving 400M+ customers, impacting **1B+ purchases** and 60% of Amazon's worldwide traffic, generating **\$xxM+** annualized financial impact and **yyK+ Prime sign-ups**.
- Designed a listwise Learning-to-Rank framework using LightGBM LambdaMART and graded relevance labels to jointly optimize payment preference and success, achieving a **20% lift in recommendation acceptance**.
- Architected a payment-decline prediction model for next-transaction payment success, reducing **Cost of Payments by \$xxM** annually and preserving **yyK+ Prime memberships**.
- Engineered a multilayer Transformer model for backup-payment decisioning, reducing unnecessary backup-payment attempts by 9% and xxxK customer contacts annually while improving customer experience.
- Designed a curriculum-learning model to classify payment-related customer contacts with limited labeled data, improving F1 score from 0.44 to 0.83 over the existing system and reducing investigation time from **2–4 hours to minutes**.
- Created a Voice-of-Customer (VoC) analytics framework using BERTopic and LLMs to identify recurring customer pain points.
- Built and scaled distributed ML batch-inference pipelines using PySpark and AWS EMR, supporting production systems.

Applied Scientist Intern

Jun 2022 – Sep 2022

- Built an Intelligent Embedding Representation to learn reusable customer embeddings for large-scale downstream ML tasks.
- Designed VAE for dimensionality reduction of tabular data, reducing **storage by 83%** and **training time by 75%**.

PUBLICATIONS

- [1] Wang, J.* , **Bhagat, L.***, et al. (2024). Spatial Steerability of GANs via Self-Supervision from Discriminator, IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**). Poster at AI4CC Workshop, Computer Vision and Pattern Recognition (**CVPR 2024**)
- [2] Kumar, S., **Bhagat, L.**, et al. (2022). Multi-Neural Network-Based Tiled 360° Video Caching with Mobile Edge Computing, Journal of Network and Computer Applications (**JNCA**), Vol. 201, Article 103342.
- [3] **Bhagat, L.**, et al. (2021). Air Quality Management Using Genetic Algorithm Based Heuristic Fuzzy Time Series Model, **The TQM Journal**, Vol. 35, No. 1, pp. 320–333.
- [4] Bansal, H., **Bhagat, L.**, Mittal, S., Tiwari, A. (2021). Image Correction and Identification of Ishihara Test Images for Color Blind Individual. IC4S, Lecture Notes in Networks and Systems, Vol. 203, Springer.
- [5] **Bhagat, L.**, Xin, W., et al. MLT-BPM: A Multilayer Dual-Transformer Approach for Optimizing Backup Payment Method, Amazon Machine Learning Conference (**AMLC 2025**) and Amazon Consumer Science Summit (**CSS 2026**).
- [6] **Bhagat, L.**, Gao, J., et al. A Curriculum Learning Approach for Customer Classification with Limited Data, Amazon Consumer Science Summit (**CSS 2026**).
- [7] **Bhagat, L.**, et al. A Clickstream-Aware Transform-Based Approach for Predicting Customer Abandonment, Workshop on Foundational Models for Tabular Data, Amazon Machine Learning Conference (**AMLC 2025**).
- [8] Gurbaxani, R., **Bhagat, L.**, et al. Optimizing Payment Recommendations with Multi-Task Learning and Mixture of Experts, Amazon Machine Learning Conference (**AMLC 2025**).
- [9] **Bhagat, L.** (2023). Parameter-Free Adversarial Attack via Learned Optimizer, Master's Thesis, University of California, Los Angeles.

* Equal contribution / co-first authorship.

PEER REVIEW & JUDGING

- **Academic Peer Review:** IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), Neural Information Processing Systems (**NeurIPS**), European Conference on Computer Vision (**ECCV**), British Machine Vision Conference (**BMVC**), and The TQM Journal.
- **Industry Peer Review:** Amazon Machine Learning Conference (**AMLC**), Amazon Computer Vision Conference (**ACVC**), and Amazon Consumer Science Summit (**CSS**).
- **Judging:** Amazon Global AI Expo and UCLA Engineering scholarship applications.

RESEARCH EXPERIENCE

Zhou Lab at UCLA

Graduate Researcher | Advisor: Bolei Zhou

Los Angeles, CA

July 2023 – Dec 2023

- Enabled **controllable generation**, object removal and zero additional annotations with no latent-direction search for GANs
- Enhanced model fidelity across multiple benchmarks, achieving **36%** reduction in **FID** scores compared to SOTA GAN baselines
- Built a **real-time inference** system integrating DragGAN for user-driven, coarse-to-fine spatial manipulation and image editing

UCLA Computational Machine Learning Lab

Master's Thesis | Advisor: Cho-Jui Hsieh

Los Angeles, CA

Oct 2022 – June 2023

- Developed a **parameter-free adversarial attack** via **meta-learning** to automate hyperparameter tuning for gradient attacks
- Evaluated TRADES-robust models, achieving an **11.3% higher attack efficacy** than PGD benchmarks on CIFAR-10

NeWS Lab, Indian Institute of Technology Hyderabad

Research Intern | Advisor: Antony Franklin

Hyderabad, India

Jun 2020 – Aug 2021

- Designed a multi-neural-network architecture within a Mobile Edge Computing framework for tiled 360° video caching
- Optimized viewport-aware content delivery using A3C, CNN-LSTM, and LFU/LRU for predictive tile-based caching
- Improved **cache hit rate by 10%**, reduced **backhaul traffic by over 35%**, and lowered end-to-end latency

Gurugram Cyber Security Cell

Cyber Security Intern | Advisor: Rakshit Tandon

Remote, India

Jun 2020 – Jul 2020

- Developed an intelligent camera application using on-device content moderation to detect and block restricted visual content
- Implemented client-side ML inference to enhance privacy and reduce security risks from cloud API transmissions

Jaypee Institute of Information Technology

Research Assistant | Advisor: Dinesh C. S. Bisht

Noida, India

Sep 2019 - May 2020

- Developed a Hybrid Adaptive Time-Variant Fuzzy Time Series model integrating Genetic Algorithms for global optimization
- Applied the model to Air Quality Index forecasting, achieving an RMSE reduction of **2 units** across two major Indian cities

AWARDS & HONORS

- **1st Place**, Amazon Payments Hackathon
- **ML Model Mastermind Award**, Amazon
- **Dive Deep Award**, Amazon
- **UCLA Graduate Council Diversity Fellowship** – Full scholarship, 1% selection rate
- **National-Level Finalist** (Top 10 in Mental Health and Psychosocial Support category), AICTE Chhatra Vishwakarma Awards 2020

INVITED TALKS & PROFESSIONAL SERVICE

- **Invited Speaker, UCLA ACM AI Research Panel** – Discussed AI research, careers, and pathways into AI/ML.
- **Panelist, Amazon Recruiting Leadership Panel** – Discussed GenAI's impact on scientific work and technical hiring.
- **Panelist, Amazon Science Intern Orientation** – Guided 100+ science interns on research execution, and career development.
- **Panelist, Amazon Seattle Intern Orientation** – Advised 500+ incoming interns on projects, networking, and internship success.
- **Technical Speaker, Amazon** – Delivered internal sessions on Generative AI and agent-based machine learning systems.

TEACHING EXPERIENCE

Machine Learning, Teaching Associate, UCLA

Fall 2023

MATLAB Programming, Teaching Associate, UCLA

Summer 2023

Advanced MATLAB Programming, Teaching Associate, UCLA

Winter 2023

SQL and Basic Data Management, Teaching Assistant, UCLA

Fall 2022

MATLAB Programming, Teaching Assistant, UCLA

Spring 2022

Cognitive Psychology, Teaching Assistant, UCLA

Winter 2022